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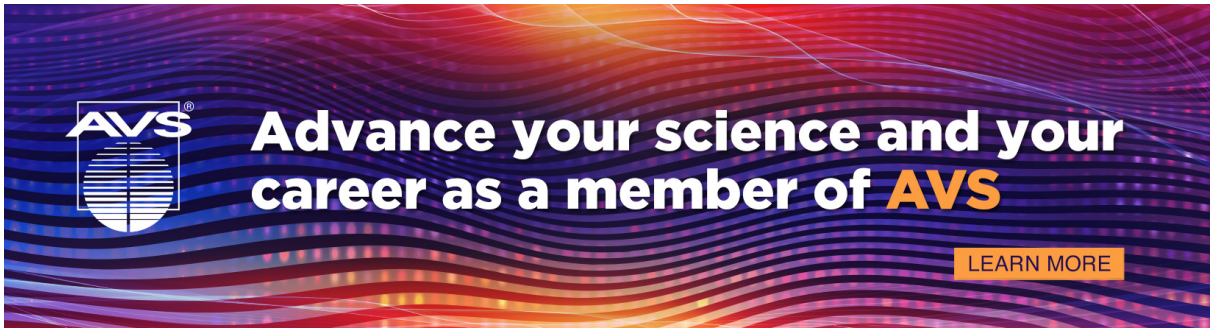
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Thermal transport and structural improvements due to annealing of wafer bonded β-Ga_2O_3|4H-SiC

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Michael E. Liao,[1,2,a] Kenny Huynh,[1] Zhe Cheng,[3] Jingjing Shi,[4] Samuel Graham,[5] and Mark S. Goorsky[1]

AFFILIATIONS

[1]Materials Science and Engineering, University of California Los Angeles, Los Angeles, California 90095
[2]U.S. Naval Research Laboratory, Washington, DC 20375
[3]Materials Science and Engineering, University of Illinois Urbana-Champaign, Urbana, Illinois 61801
[4]Mechanical Engineering, Georgia Institute of Technology, Atlanta, Georgia 30332
[5]Mechanical Engineering, University of Maryland, College Park, Maryland 20742

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[a]**Author to whom correspondence should be addressed:** meliao@ucla.edu

ABSTRACT

The impact of postbond annealing on the structural and thermal characteristics of 130 nm thick exfoliated (2̄01) β-Ga_2O_3 (via H^+ ion implantation) wafer bonded to (0001) 4H-SiC was studied. Thirty nanometer amorphous-Al_2O_3 was grown on the β-Ga_2O_3 substrates prior to bonding as an interlayer between β-Ga_2O_3 and 4H-SiC. The surface activated bonding technique was utilized for bonding, which induces a thin nanometer amorphous interfacial region at the bonded interface (Al_2O_3|4H-SiC). We demonstrate annealing the bonded structure at 800 °C up to 1 h is beneficial: (1) the removal of residual strain in the exfoliated β-Ga_2O_3 layer that was due to the exfoliation implant, (2) reduction of lattice mosaicity in the β-Ga_2O_3 layer, (3) nearly complete recrystallization of the amorphous bonded interfacial region, and (4) partial recrystallization of the initially amorphous-Al_2O_3 interlayer. The thermal characteristics correspondingly improve with the improvement in structural characteristics. The thermal conductivity of the as-bonded β-Ga_2O_3 layer was 2.9 W/m K, and the thermal boundary conductance of the bonded interface was 66 MW/m^2 K. After annealing at 800 °C for 1 h, triple-axis x-ray diffraction ω:2θ measurements showed a reduction in strain for the β-Ga_2O_3 layer and the symmetric (2̄01) rocking curve widths. We simultaneously observe a doubling of the β-Ga_2O_3 thermal conductivity to 6.0 W/m K and a 20% increase in the thermal boundary conductance. However, upon further annealing up to 10 h and fully recrystallizing both the Al_2O_3 interlayer and bonded interface, the thermal boundary conductance dropped by ~30%. This preliminary result suggests that crystalline heterointerfaces may not necessarily be the most optimal interfacial structure for thermal transport.

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I. INTRODUCTION

β-Ga_2O_3 is widely recognized as a promising semiconductor for next generation high-power devices due to its ultrawide bandgap and high electric breakdown field.[1–3] Recent reports of β-Ga_2O_3 device fabrication demonstrate the realization of its potential for electronic devices.[4–6] However, one of the most critical challenges of β-Ga_2O_3 is its very low thermal conductivity ranging from 11 to 27 W/m K,[7] which is one to two orders of magnitude lower than other semiconductors such as Si, GaN, 4H-SiC, and diamond.[1] Low thermal conductivity poses an issue for β-Ga_2O_3 power devices due to inefficient heat dissipation. It has been documented that poor heat management in either lateral or vertical β-Ga_2O_3 devices degrades device performance[8] and can even cause permanent device failure.[9,10] It is evident that heat generated during β-Ga_2O_3 device performance requires additional thermal management. While heat-related issues are recognized in the current literature, the understanding of thermal management for β-Ga_2O_3 is still in its infancy and much less studied than the electrical characteristics of devices.

Theoretical studies for thermal management of β-Ga_2O_3 devices have investigated various cooling methods: bottom-side

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cooling (substrate side), top-side cooling (device side), and double-side cooling.[11–13] It was found that, especially for vertical devices (where reducing the β-Ga_2O_3 thickness is not an option), either top-side or double-side cooling is expected to be most effective for thermal management. While there are some experimental reports studying possible thermal management solutions,[14–19] a fundamental understanding of thermal transport across interfaces is still lacking. One important interface in various lateral and vertical device structures is the semiconductor–dielectric interface. These technologically relevant interfaces are utilized, for example, in metal–oxide–semiconductor field effect transistors (MOSFETs) and metal–oxide–semiconductor capacitors (MOSCAPs). One top-side cooling approach could be employed through these semiconductor–dielectric interfaces The focus of this work is the thermal transport across wafer bonded (2̄01) β-Ga_2O_3|dielectric|(0001) 4H-SiC interfaces, which could be applied to either bottom-side or top-side cooling approaches. For bottom-side cooling, a thin β-Ga_2O_3 layer would be bonded to a high thermal conductivity substrate such as 4H-SiC, possibly using an intermediate dielectric layer. Top-side cooling would instead require transferring a thin layer of high thermal conductive material on the dielectric materials. A recent study successfully demonstrated working β-Ga_2O_3 MOSFET devices on a 4H-SiC-based composite wafer and reduced the channel temperature by ∼56% via bottom-side cooling.[20] Even further improvements to thermal management are expected if top-side cooling was also employed in parallel with bottom-side cooling, i.e., the double-side cooling approach.

A dielectric layer in this current study used to assist the bonding between β-Ga_2O_3 and 4H-SiC is ∼30 nm Al_2O_3, which is comparable to the thicknesses of Al_2O_3 used in various β-Ga_2O_3 MOSFETs,[20–24] MOS diodes,[25,26] and MOSCAPs[27–30] that range from ∼10 to ∼60 nm. A continuation of our previous work,[31] this current study provides further structural analysis of the heterointerfaces for this β-Ga_2O_3|Al_2O_3|4H-SiC system. We provide (1) a more objective and quantitative approach in assessing the amorphous/crystalline character of the bonded interface by examining the diffuse scatter intensity of the interface, (2) a documentation of the intermediate stages of the Al_2O_3 interlayer recrystallization process and its corresponding thermal transport properties, and (3) a completely recrystallized 4H-SiC bonded interface prepared with the surface activated bonding technique, which to the best of our knowledge has yet been reported in the current literature. Here, a 130 nm thick exfoliated (2̄01) β-Ga_2O_3 layer was wafer bonded to a (0001) 4H-SiC substrate using a thin ∼30 nm Al_2O_3 interlayer at the bonded interface. β-Ga_2O_3 and 4H-SiC are aligned in-plane such that [010] β-Ga_2O_3||[11$\bar{2}$0] 4H-SiC. With this alignment of the zone axes, β-Ga_2O_3 and 4H-SiC have the lowest twist misalignment (i.e., the least amount of lattice mismatch). Theoretical modeling suggests that twist misalignment across interfaces increases the thermal boundary resistance.[32] Previous work on wafer bonded InP|InP demonstrated that twist misalignment hinders electronic transport and increases the interfacial electrical resistance.[33] The same trend was demonstrated for electronic transport across wafer bonded GaAs and InP with intentionally induced tilt misalignment.[33–35] In general, misalignment across interfaces impedes the transport of phonons and electrons. By studying the structural and thermal characteristics of this low-mismatch scenario, we aim to leverage further fundamental understanding of more technologically relevant orientations, which include (010) and (001) β-Ga_2O_3. While the structure fabricated in this study closely mimics a lateral device structure for bottom-side cooling, the characteristics of the interfaces themselves are expected to be applicable in the vertical device setting for top-side cooling as well. In this scenario, a thin layer of 4H-SiC would be exfoliated and transferred[36–40] on a β-Ga_2O_3 substrate with a dielectric surface layer.

II. EXPERIMENT

50 mm diameter (2̄01) β-Ga_2O_3 substrates grown using the edge-defined film-fed growth method were first implanted with H^+ ions at 35 keV with a dose of 1×10^{17} cm^{-2} at room temperature. Then, the 30 nm Al_2O_3 layer was deposited on the implanted β-Ga_2O_3 substrate surfaces via plasma-enhanced atomic layer deposition at 200 °C. While our previous work demonstrated reducing the Al_2O_3 layer from 30 to 10 nm increases the thermal boundary conductance,[31] 30 nm is used in this current work as a middle-ground system between the thinnest and thickest dielectric layers currently used in devices. The processed β-Ga_2O_3 substrates were subjected to an ion sputtering treatment consisting of Ar and Si ions to create dangling bonds under vacuum (5×10^{-6} Pa); 4-in. (0001) 4H-SiC substrates were also subjected to the same ion sputtering surface treatment in the same bonding chamber. The two surface treated materials were then brought face-to-face and bonded at room temperature.[31] The resulting bonded interface was a 2.9 nm thick amorphous region between Al_2O_3 and 4H-SiC. The thin amorphous bonded interface is a common occurrence when bonding covalent materials using this ion bombardment surface treatment.[41–46] The bonded structure was then annealed at 450 °C for 6 h to induce H_2 bubble growth at the projected range within the β-Ga_2O_3 substrate and exfoliated a ∼400 nm thick layer across the entire 50 mm wafer area. The bonded β-Ga_2O_3 layer was then polished to remove the surface roughness from exfoliation, resulting in a final thickness of ∼130 nm bonded to 4H-SiC with an Al_2O_3 interlayer at the bonded interface.[31] The thickness variation across the entire 50 mm transferred and polished film was ∼2%, which confirms the large-scale compatibility of achieving large-area β-Ga_2O_3 films transferred to other substrates. Postbond annealing was done at 800 °C for up to 10 h in an ambient atmosphere, using a 5 °C/min ramp up rate on an Instec high temperature stage in an x-ray diffractometer. The peak shift of the (0008) 4H-SiC reflection was used to verify the bonded sample reached 800 °C by using the known coefficient of thermal expansion for 4H-SiC.[47]

The thermal conductivity of the exfoliated β-Ga_2O_3 and thermal boundary conductance (TBC) before and after the postbond anneal were measured using time-domain thermoreflectance (TDTR).[48,49] The pump radius used was 10.1 μm in areal diameter, while the probe radius was 5.8 μm in areal diameter. A low modulation frequency of 2.2 MHz was employed to extract the TBC of the bonded interface buried beneath the exfoliated β-Ga_2O_3 layer.[31] Structural characterization was performed with a high-resolution Bruker-JV D1 x-ray diffractometer using triple-axis diffraction (acceptance angle of ∼10″). The incident x-ray beam is conditioned by a Göbel mirror[50] and a (220) channel-cut silicon crystal, which produces a highly collimated monochromatic beam of Cu $K\alpha_1$

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radiation. The scattered beam optics is a four-bounce (220) channel-cut silicon analyzer crystal. Triple-axis symmetric ω:2θ and ω scans were measured to measure strain and lattice tilt, respectively. An FEI Nova 600 DualBeam focused ion beam system was used to prepare transmission electron microscopy samples. An FEI TITAN S/TEM operating at a 300 keV accelerating voltage was then employed to generate scanning transmission electron microscopy (STEM) images aligned to the [010] β-Ga_2O_3 and [11$\underline{2}$0] 4H-SiC zone axes.

III. RESULTS AND DISCUSSION

The cross-sectional bright-field scanning transmission electron microscopy (STEM) image of the bonded β-Ga_2O_3|Al_2O_3|4H-SiC structure is shown in Fig. 1. The Al_2O_3 intermediate layer is ~30 nm thick, and the exfoliated β-Ga_2O_3 thin film is ~130 nm thick. Note that the bonded interface is between the Al_2O_3 interlayer and 4H-SiC substrate. Magnified high-resolution TEM images of the bonded interface for the as-bonded state and after annealing 800 °C for 1 h are shown in Figs. 2(a) and 2(b), respectively. The fast Fourier transform (FFT) in Fig. 2(c) of the as-bonded state shows that the ~2.9 nm layer at the bonded interface is amorphous SiC. After annealing, the FFT in Fig. 2(d) shows reciprocal lattice points that indicate the amorphous SiC recrystallized during anneal. For a semiquantitative comparison, the FFT for bulk 4H-SiC is shown in Fig. 2(e), and the integrated line extractions taken along the horizontal (Q_X axis) for all the FFTs are plotted in Fig. 2(f). The FFT for bulk 4H-SiC is measured within the substrate away from the bonded interface where the 4H-SiC lattice is undistorted. For the as-bonded state, SiC is amorphous and correspondingly shows the greatest amount of diffuse scatter intensity between ±6 nm^{-1} along Q_X. After annealing, the diffuse scatter intensity reduces toward the same intensity of bulk 4H-SiC, and the reciprocal lattice points corresponding to the (10$\underline{1}$0) 4H-SiC planes emerged.

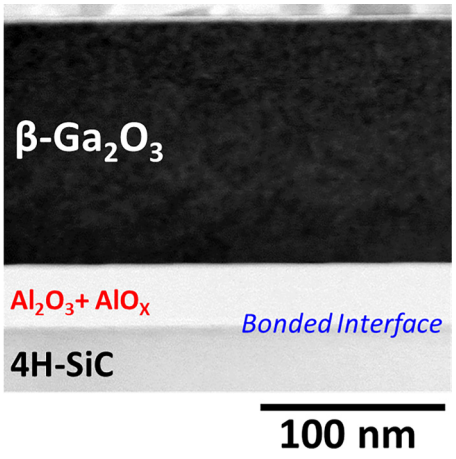


FIG. 1. Cross-sectional bright-field STEM image of the wafer bonded β-Ga_2O_3|Al_2O_3|4H-SiC. The bonded interface is between the Al_2O_3 layer and the 4H-SiC substrate.

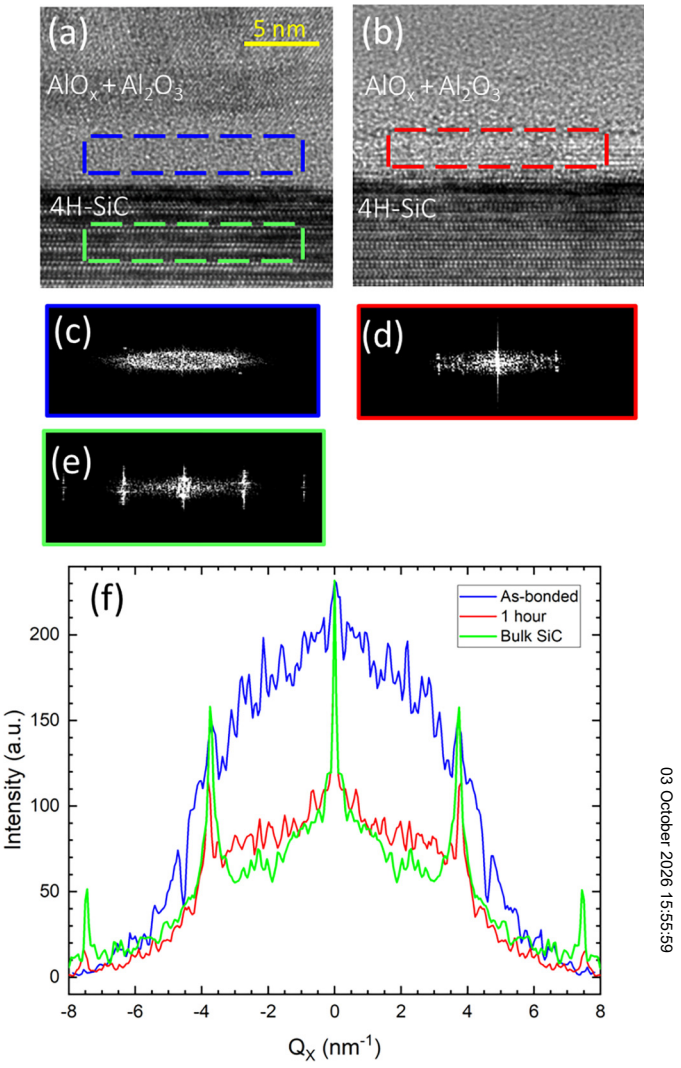


FIG. 2. Cross-sectional high-resolution transmission electron microscopy images of the wafer bonded (2$\underline{0}$1) β-Ga_2O_3|(0001) 4H-SiC: (a) as-bonded and (b) 1 h anneal at 800 °C. FFTs for each of the boxed areas are shown for (c) amorphous SiC, (d) recrystallized 4H-SiC, and (e) bulk 4H-SiC beneath the bonded interface. The extracted integrated line scans of the fast Fourier transform images are shown in (f). Satellite peaks in (f) correspond to the (10$\underline{1}$0) 4H-SiC interplanar spacing.

During the anneal, the alumina interlayer also recrystallized simultaneously with 4H-SiC at the bonded interface. Initially, the alumina layer is amorphous, as shown in the cross-sectional TEM image and the lack of reciprocal lattice points in its respective FFT shown in Fig. 3(a). After annealing at 800 °C for 1 h, approximately half of the thickness of the alumina layer recrystallized starting at the β-Ga_2O_3 interface, while the other half of the interlayer adjacent to the 4H-SiC interface remained amorphous as shown in Fig. 3(b). The crystallized Al_2O_3 layer is (0001) textured, with some

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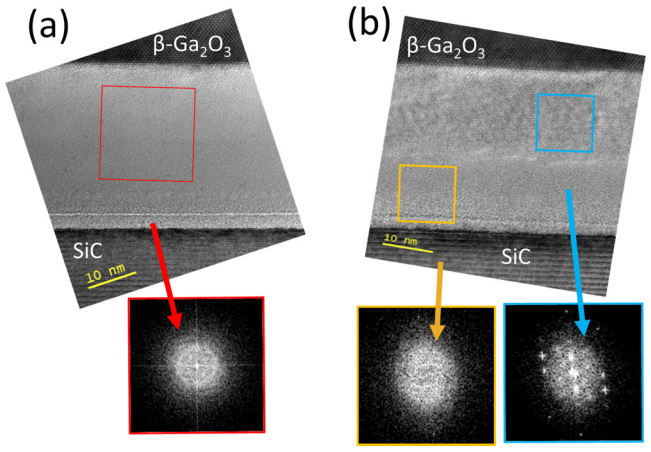


FIG. 3. Cross-sectional high-resolution transmission electron microscopy images of the alumina interlayer: (a) as-bonded and (b) 1 h anneal at 800 °C. Recrystallization of the Al_2O_3 initiates at the (2̄01) β-Ga_2O_3 interface and the crystallization front propagates toward the 4H-SiC.

regions having an in-plane relationship [010] β-Ga_2O_3‖[11$\bar{2}$0] Al_2O_3 shown in Fig. 3(b), while other regions having an in-plane relationship [010] β-Ga_2O_3‖[10$\bar{1}$0] Al_2O_3 (not shown).

The recrystallization of the 4H-SiC layers at the bonded interface and Al_2O_3 layer increases the thermal boundary conductance (TBC) by ~20% compared to the as-bonded state. The contribution to the conductance from the crystallized Al_2O_3 is consistent with work that showed the thermal conductivity of crystalline-Al_2O_3 is higher than its amorphous counterpart.[51] The as-bonded state exhibited a TBC of 66 MW/m^2 K, while after annealing at 800 °C for 1 h, the TBC increased to 77 MW/m^2 K. The TBC reported here combines the contributions from (1) the β-Ga_2O_3|Al_2O_3 interface, (2) the Al_2O_3 interlayer, and (3) the Al_2O_3|4H-SiC interface. Xu *et al.*[52] studied a similar β-Ga_2O_3|Al_2O_3|4H-SiC structure except used a thinner 20 nm Al_2O_3 intermediate bonding layer and annealed at an even higher temperature of 900 °C to achieve a higher TBC of ~130 MW/m^2 K. However, annealing at lower temperatures may be an important consideration for systems with limited thermal budgets. Song *et al.*[53] point out that a thermal boundary conductance >17 MW/m^2 K would be sufficient to reasonably manage the heat generated from a β-Ga_2O_3 transistor operating at a 10 W/mm power density that would exceed 1500 °C without heat dissipation strategies. Hence, even the TBC of the as-bonded state would satisfy this requirement. Annealing may be employed to further improve the TBC to comply with heat dissipation requirements for devices operating at even higher power densities. While reducing the intermediate Al_2O_3 layer is one way to increase TBC, it may not always be an available parameter to tune for device structures where a dielectric layer (e.g., Al_2O_3) is needed to electrically isolate the transferred β-Ga_2O_3 from the underlying substrate. For example, employing thicker dielectric layers would reduce leakage current.

The TBC values achieved here are superior to bonded structures with relatively weak van der Waals forces, e.g., β-Ga_2O_3 on diamond interfaces fabricated using the mechanical tape exfoliation and transfer technique (TBC ranges from ~8 to ~17 MW/m^2 K).[17,54] Polycrystalline β-Ga_2O_3 on diamond has already been demonstrated to exhibit high TBC values ranging from ~140 to ~180 MW/m^2 K.[18] However, despite achieving high TBC values, the thermal conductivity of polycrystalline films is always lower than single crystal films for a given thickness due to grain-boundary phonon scattering. Likewise, while epitaxial β-Ga_2O_3 grown on SiC[55] and sapphire[56] have been demonstrated to have TBC values ranging from ~140 to as high as ~500 MW/m^2 K, the films have a high density of stacking faults (on SiC[55]) and twinning (sapphire[56]). These defects would increase phonon scattering and decrease the β-Ga_2O_3 film thermal conductivity. Thus, direct wafer bonding and transferring high quality β-Ga_2O_3 is a way to avoid heteroepitaxy-related defects in order to achieve both high TBC and high film thermal conductivity. For example, direct wafer bonding β-Ga_2O_3 on diamond is speculated to exhibit at least comparable TBC values as the polycrystalline β-Ga_2O_3 films on diamond while, at the same time, achieve higher single-crystalline film thermal conductivity. The TBC values for some wafer bonded heterostructures in units of MW/m^2 K are 60–130 for β-Ga_2O_3|4H-SiC,[31,52] 140 for GaN|Si,[46] 170–230 for GaN|4H-SiC,[57] 50–90 for GaN|diamond,[58] 250 for GaN|BAs,[59] and 35 for Si|Ge.[60] Some of the interfaces of these structures are even on the same level as heterostructures fabricated through growth methods. While the direct wafer bonding of β-Ga_2O_3 to diamond has been demonstrated,[19] the thermal transport characterization of this interface had not yet been reported.

Triple-axis x-ray diffraction symmetric (201) β-Ga_2O_3 ω:2θ scans are shown in Fig. 4(a). The (0004) 4H-SiC reflection corresponds to 0″ along the ω:2θ scanning axis for both as-bonded and postanneal scans. After annealing, the (2̄01) β-Ga_2O_3 peak shifts by ~100″ toward higher angles, which corresponds to a reduction of ~0.3% in strain. For the as-bonded state (i.e., postexfoliation), the peak being at a lower Bragg angle than bulk β-Ga_2O_3 prior to annealing indicates that the film is in a tensile strain state. Even after the exfoliation, there may be residual hydrogen dissolved and/or intercalated in the β-Ga_2O_3 film or Ga and O point defects. The concentration of residual H based on Stopping and Range of Ions in Matter (SRIM)[61] simulations is on the order of ~1×10^{21} cm^{-3}, i.e., the H concentration at the depth from the surface that matches the final thickness of the transferred and polished film (compared to the peak concentration of ~7×10^{21} cm^{-3} at the implant projected range). This strain is removed after annealing at 800 °C for 1 h, and the lattice reaches the value expected for bulk β-Ga_2O_3. The thermal conductivity of the β-Ga_2O_3 film increased after the anneal from 2.9 to 6.0 W/m K. The reduction in the thermal conductivity due to strain-field induced phonon-defect scattering discussed in our previous work.[31] The thermal conductivity value here, however, is lower than the bulk value (~13 W/m K) due to thickness effects. Increasing the thickness of a film (~nm to ~μm) increases the thermal conductivity of the film.[31,53,56,62–64] The expected thermal conductivity for 130 nm of (2̄01) β-Ga_2O_3 is ~7 W/m K,[31] which is comparable to our results. The triple-axis symmetric (2̄01) β-Ga_2O_3 rocking curves in Fig. 4(b) show the FWHM decreases from 120″ as-bonded to 70″ after the anneal. The reduction in lattice mosaicity (i.e., reduction in peak width) and corresponding improvement in thermal

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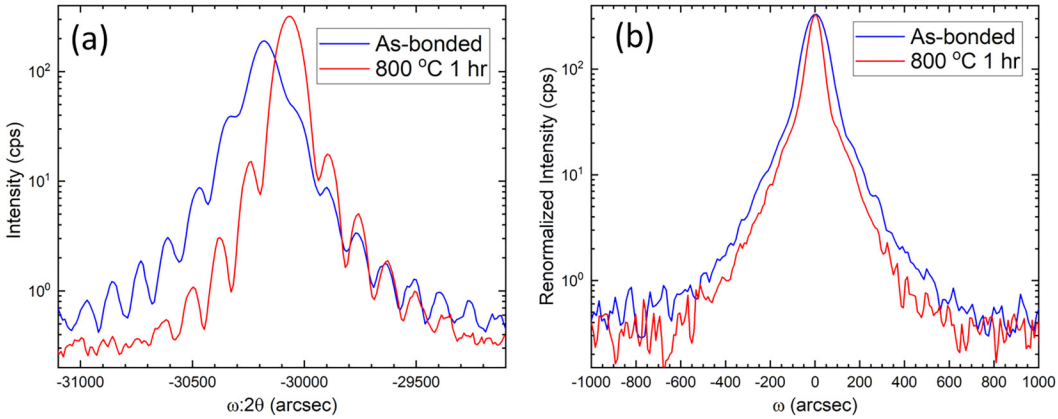


FIG. 4. Triple-axis x-ray diffraction: (a) ω:2θ and (b) ω of the symmetric (201) β-Ga_2O_3 layer. After annealing for 1 h, residual strain from the ion implantation was reduced and the rocking curve FWHM decreased from 120″ to 70″.

conductivity is consistent with the dissolution of implant-related extended defects due to annealing.[65] A study on the impact of irradiation and subsequent annealing on the electrical characteristics of the β-Ga_2O_3 is underway. Recent work on β-Ga_2O_3 Schottky diodes showed that upon hydrogen implantation, both the leakage current and on current decreased.[66] The effect of annealing after irradiation has not yet been studied. We speculate that upon annealing, the film would recover toward its original electrical transport properties based on work done on a different material system. It has been shown that the carrier lifetime in 4H-SiC can be reduced by implanting with hydrogen, and upon annealing, the carrier lifetime increases toward its original preimplanted value.[67]

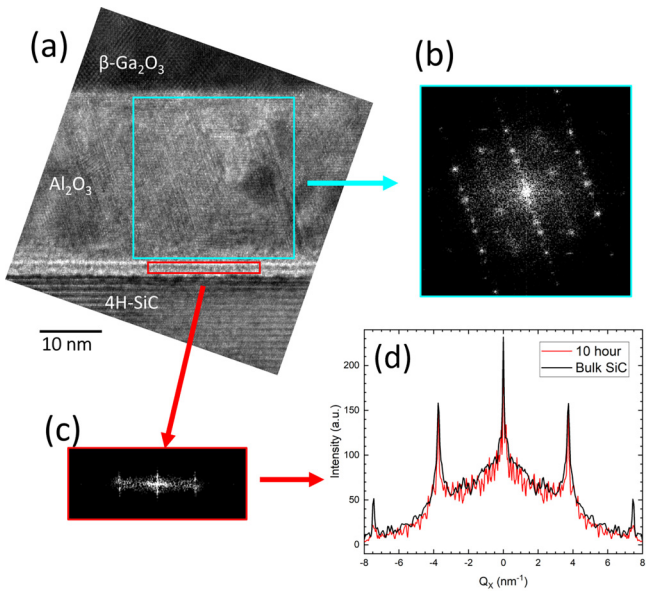


FIG. 5. (a) Cross-sectional high-resolution transmission electron microscopy images after annealing for 10 h anneal at 800 °C. Complete recrystallization of the bonded region was achieved: fast Fourier transform images of (b) the Al_2O_3 layer showing distinct reciprocal lattice points characteristic of a crystalline layer and (c) 4H-SiC at the bonded interface. (d) The extracted integrated line scans for the 4H-SiC show the complete recrystallization after anneal.

Further annealing up to 10 h at 800 °C was performed, and complete recrystallization was achieved for both the Al_2O_3 interlayer and 4H-SiC at the bonded interface as shown in Fig. 5. When compared to Fig. 2(f), it is observed that the slight diffuse scatter present in the 4H-SiC at the bonded interface after annealing for 1 h at 800 °C is removed after annealing for 10 h at 800 °C. Bonding and annealing 4H-SiC using this ion bombardment approach has been reported in the literature,[31,42,44,52,68] but no report to the best of our knowledge has demonstrated complete 4H-SiC recrystallization at the bonded interface. Preliminary TDTR measurements show that the TBC after complete recrystallization decreased to ~55 MW/m^2 K, which is a ~30% decrease compared to the value measured after the 1 h anneal. This suggests that having crystalline-to-crystalline interfaces may not necessarily have the most optimal TBC. Theoretical calculations by Gordiz and Henry[69] suggest that for Si and Ge, having crystalline–amorphous interfaces could lead to improved thermal transport compared to only crystalline–crystalline interfaces. Giri *et al.*[70] compared experimental data and found that either amorphous|amorphous or crystalline|amorphous interfaces generally exhibit higher TBC values than crystalline|crystalline interfaces whose materials have a large elastic moduli mismatch. Interfaces with at least one side amorphous despite how large the elastic moduli mismatch can be just as thermally conductive as, if not greater than, crystalline|crystalline interfaces with low elastic moduli mismatch.[70] Here, we found experimentally that the following combination of interfaces corresponds to the optimized TBC for this bonded system: crystalline-β-Ga_2O_3|crystalline-Al_2O_3, crystalline-Al_2O_3|amorphous-Al_2O_3, and amorphous-Al_2O_3|crystalline-4H-SiC. We speculate that the phonon density of states overlap between this combination of crystalline and amorphous interfaces is

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superior over the density of states overlap between only crystalline interfaces. Last, the β-Ga_2O_3 film rocking curve width after annealing for 10 h increased to 80″ and the film thermal conductivity correspondingly decreased to 4.9 W/m K. Both of these values are intermediate to the as-bonded and 1 h annealed state (120″, 2.9 W/m K and 70″, 6.0 W/m K, respectively). Degradation studies of these single crystal β-Ga_2O_3 films with annealing temperature and duration are underway.

IV. SUMMARY AND CONCLUSIONS

Successful exfoliation and bonding of (2̄01) β-Ga_2O_3 on (0001) 4H-SiC using a 30 nm alumina interlayer was demonstrated. The thermal transport and structural characteristics of the amorphous 4H-SiC|Al_2O_3 bonded interface, Al_2O_3 interlayer, and β-Ga_2O_3 film were assessed. The amorphous bonded interface was shown to recrystallize along with the Al_2O_3 interlayer, which improved the TBC by ~20% to 77 MW/m^2 K. Residual strain in the β-Ga_2O_3 film from the ion implantation process was simultaneously removed and improved the thermal conductivity of the film from 2.9 to 6.0 W/m K. While the surface activated bonding technique is useful for bonding dissimilar materials, the amorphous or lattice-damaged interfaces created by this process can be engineered (e.g., annealing) to alter interfacial transport properties. Preliminary thermal analysis shows that fully recrystallizing the Al_2O_3 interlayer corresponds to a drop in the overall TBC of the bonded structure, which suggests that having completely crystalline interfaces may not necessarily achieve the most optimal thermal transport properties for a given material combination.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Michael E. Liao: Conceptualization (equal); Data curation (equal); Formal analysis (equal); Investigation (equal); Methodology (equal); Visualization (equal); Writing – original draft (lead); Writing – review & editing (equal). **Kenny Huynh:** Conceptualization (supporting); Data curation (supporting); Formal analysis (supporting); Investigation (supporting); Methodology (supporting); Visualization (supporting). **Zhe Cheng:** Data curation (supporting); Validation (equal). **Jingjing Shi:** Data curation (supporting); Validation (equal). **Samuel Graham:** Funding acquisition (equal); Project administration (equal); Resources (equal); Supervision (equal). **Mark S. Goorsky:** Funding acquisition (equal); Project administration (equal); Supervision (equal); Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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